MATHEMATICS II NOTES

Unit 1: Differential Equations

1. Definition: A differential equation is an equation involving derivatives of a function.

2. Order and Degree of Differential Equation.

3. First Order Differential Equations – Variable separable method.

4. Linear Differential Equations.

5. Applications of Differential Equations.

Unit 2: Laplace Transform

1. Definition of Laplace Transform.

2. Properties of Laplace Transform.

3. Inverse Laplace Transform.

4. Application in solving differential equations.

Unit 3: Fourier Series

1. Periodic Functions.

2. Fourier Series Formula.

3. Half Range Sine and Cosine Series.

Unit 4: Partial Differential Equations

1. Formation of PDE.

2. Solution of First Order PDE.

3. Method of Separation of Variables.

Unit 5: Complex Numbers

1. Complex number definition.

2. Euler’s Formula.

3. De Moivre’s Theorem.